BILAWAL HUSSAIN

Software Engineer | ASP.NET Core, C#, Angular, SQL

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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of enterprise experience building scalable applications using ASP.NET Core, C#, Angular, PostgreSQL, and SQL Server. Skilled in developing RESTful APIs, implementing business workflows, designing relational databases, and contributing to microservices-based Manufacturing Execution Systems (MES). Comfortable across backend and frontend layers, with hands-on experience in API integrations, production troubleshooting, and delivering reliable software solutions.

TECHNICAL SKILLS

Languages:	C#, TypeScript, JavaScript, SQL
Backend:	ASP.NET Core, RESTful APIs, Razor Pages, LINQ, EF Core, ADO.NET, JWT
Frontend:	Angular, HTML5, CSS3, jQuery
Databases:	SQL Server, PostgreSQL
Architecture & Tools:	Clean Architecture, Microservices, RabbitMQ, Docker, Azure DevOps, Git, Google OAuth

PROFESSIONAL EXPERIENCE

Software Engineer (.NET & Angular) | **NKU Technologies** — **Jul 2025 – Present | Lahore, Pakistan**

- Developed and maintained backend services and RESTful APIs for O3Sigma and Mesonex MES using ASP.NET Core and C#.
- Developed and maintained Angular frontend components, integrated REST APIs, and resolved UI issues across MES modules.
- Designed and optimized PostgreSQL/SQL Server database objects, queries, functions, and data migrations.
- Built microservices integrations using API Gateway, RabbitMQ, Docker, and service-to-service communication.
- Implemented core MES workflows: production planning, execution, production losses, palletization, OEE, NCR, machine/product configuration, and multi-plant operations.
- Developed a Procurement data warehouse for TrueBook ERP using star schema and RabbitMQ.
- Owned Honest Abe Roofing platform end-to-end, delivering roofing options, pricing, ROI, repair/replacement, protection plans, and homeowner presentations.

Technologies: ASP.NET Core, C#, Angular, PostgreSQL, SQL Server, RabbitMQ, Docker, Microservices

ASP.NET Core Developer | **Evonicsoft** — **Sep 2024 – Jun 2025 | Lahore, Pakistan**

- Developed and maintained client-facing websites using Umbraco CMS and .NET.
- Built reusable CMS components and content structures to simplify content management.
- Implemented website features and customized Umbraco functionality based on requirements and designs.
- Debugged and resolved issues across development and testing to ensure expected functionality.

Technologies: ASP.NET Core, C#, Umbraco CMS, .NET, SQL Server, JavaScript

KEY PROJECTS

O3Sigma
Industrial Desktop + Cloud Application

- Contributed to a multi-plant OEE monitoring platform with a cloud portal and an edge desktop application, both built on Angular and .NET Core.
- Implemented plant onboarding and master data setup: departments, machines, products, machine-product mapping, NCR limits, shift timings, checklists, downtime, and waste categories.
- Built .NET Core APIs to process real-time machine data (via OPC) stored in PostgreSQL and calculate OEE, Performance, Quality, Availability, production losses, downtime, and waste.
- Developed Angular dashboards for OEE, Pareto analysis, NCR management, and checklists, along with production order handling.

  Tech: ASP.NET Core, Angular, PostgreSQL, Docker, OPC, Electron
  Role: Cloud-side and edge-side backend API and Angular frontend development.

Mesonex
Manufacturing Execution System (MES) Platform

- Worked as a .NET Core API and Angular developer on a manufacturing execution system platform.
- Calculated downtime and waste metrics and visualized them through Angular charts.
- Computed historical OEE and presented it through multiple dashboards and chart views.
- Implemented palletization workflows: lot number assignment, barcode generation, pallet packing quantities, and delivered vs. remaining tracking.
- Built a production summary dashboard showing pending, in-progress, and completed orders per plant, with estimated completion dates based on downtime, waste, and delivery estimates.

  Tech: ASP.NET Core, Angular, SQL Server, RabbitMQ
  Role: Backend API and Angular frontend development.

Honest Abe Roofing
Sales Quoting & Job Management Platform

- End-to-end roofing sales and job workflow platform for a US-based roofing company, built with ASP.NET Core Razor Pages (tablet-first, Google sign-in, role-based).
- Implemented Roofing Advisor workflow: check-in with live location, authorized photo capture, leakage measurements, 16-question inspection, repair material suggestion, and replacement design wizard (material, style, color, add-ons).
- Built ROI comparison, payment/financing flow, e-signature, office handoff to production, and job close-out with final e-sign.
- Delivered protection plans and homeowner presentation features as part of the sales workflow.
- Integrated CRM APIs (data fetch and post) for customer, job, and media records.

  Tech: ASP.NET Core Razor Pages, CRM APIs, Google OAuth
  Role: End-to-end ownership (design, backend, workflow, and delivery).

TrueBook ERP
Procurement Data Warehouse

- Designed and built a procurement data warehouse using star schema (fact and dimension tables) covering PR, quotations, PO, GRN, purchase returns, and reports.
- Populated warehouse tables initially via RabbitMQ pipelines, then built ETL jobs for ongoing data sync.
- Created reporting layer to retrieve procurement insights from the warehouse.

  Tech: SQL Server, RabbitMQ, ETL, Star Schema
  Role: Data warehouse design, ETL development, and reporting.

EDUCATION

BS Computer Science | Gomal University (Govt. College No.1, Dera Ismail Khan) **2019 – 2023**

Intermediate in Computer Science (ICS) | Govt. College No.1, Dera Ismail Khan **2017 – 2019**

Matriculation | Govt. Higher Secondary School No.2, Dera Ismail Khan **2015 – 2017**

LANGUAGES

English: Intermediate **Urdu:** Native